AI-Driven Technical Change, the Real Interest Rate, and the Real Wage: Closed Forms in a Production Economy with Adjustment Costs

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Abstract

This note uses Kiley (2004) (working paper Kiley, 2000) as a blueprint for analyzing AI-driven technical change. The blueprint has one capital stock and two pairings of production and capital-accumulation technologies, each with a closed form. (i) CES production with an investment adjustment cost that vanishes on the balanced growth path gives $Q = 1$ and closed forms for the interest rate, the wage, the labor share and value/output on that path. (ii) CES production with a CES capital-accumulation function of the same elasticity, $K'^{\rho} = \bar{A}^{\rho}[(1-\delta)K^{\rho} + \delta I^{\rho}]$, gives an exact closed form for the *entire* equilibrium path when the intertemporal elasticity of substitution equals the production elasticity. The saving rate is then constant, $s = [B\bar{A}^{\rho}\delta a_K/(1 - B\bar{A}^{\rho}(1-\delta))]^{\varepsilon}$, and capital follows a linear law of motion in K^{ρ}. The Cobb-Douglas case, $K' = \bar{A}K^{1-\delta}I^{\delta}$ with log utility, is the $\varepsilon \to 1$ limit. There value/output is constant and the real interest rate and real wage follow from a log-linear law of motion. I add AI capital as a third input. It combines with the capital-labor bundle through a CES aggregator, and its technology improves either disembodied (a factor Γ on the installed stock) or embodied (a falling price P_A of new units, as in Greenwood, Hercowitz and Krusell, 1997). Under pairing (i), the balanced growth path has closed forms for any elasticity, an AK regime for substitutes and a Baumol limit for complements. Under pairing (ii), with all three inputs sharing one elasticity, the transition paths of r and W after any AI shock are closed form. Saving rates depend only on the technology path, not on the state. Embodied and disembodied progress with the same long-run effect raise $\sum_t \ln(1 + r_{t+1})$ by exactly the same amount, the long-run log output gain, and differ only in timing. A gradual (S-curve) disembodied shock raises r sooner and more. A surprise permanent step does the reverse: embodied progress raises the ex-ante real rate more, because disembodied progress delivers its first gain as a windfall. With complements, a disembodied step even lowers r.

1 The blueprint: two closed-form pairings with one capital stock

Kiley (2000) studies a representative-agent economy (notation as in that paper). The household maximizes

$$U = E_t \sum_{j=t}^{\infty} B^{j-t} \frac{[C(j)^b L(j)^{1-b}]^{1-s}}{1-s}, \qquad L = 1 - N, \tag{2.1}$$

and holds claims to capital, $C(t) + Q(t)K(t+1) = (Q(t) + D(t))K(t) + W(t)N(t)$. The calibration (Table 1 of the working paper) is $s = 1$ (log utility), $b = 0.25$, $a = 0.282$, $d = 0.066$, $r = 0.075$, with $c = 15$ for "moderate" adjustment costs. Let $\rho \equiv 1/B - 1$ be the rate of time preference and

$\theta \equiv 1-b(1-s)$ the inverse intertemporal elasticity (footnote 11 of the paper). With constant hours on a balanced growth path, $1+r=(1+g)^{\theta}/B$.

I could not obtain the published article's text. The working paper has the Cobb-Douglas, quadratic-cost case. The pairings below follow the published article's structure as the author describes it. CES production pairs with an adjustment-cost function, and with a CES investment/accumulation function of the same elasticity, which gives an exact solution with a constant saving rate. The Cobb-Douglas production/accumulation pair is the special case $\varepsilon = 1$. I derive the closed forms here.

1.1 Pairing (i): CES production and adjustment costs

Production and the adjustment cost are

$$Y = \left[\alpha^{1/\varepsilon} K^{\frac{\varepsilon-1}{\varepsilon}} + (1-\alpha)^{1/\varepsilon}(ZN)^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}} - \Psi[I/K]K, \qquad (2.7')$$
$$\Psi[I/K]K = 0.5\, c\,(I/K - x)^2 K, \qquad x = g + d,$$

with $K(t+1) = I(t) + (1-d)K(t)$. (The paper writes the cost as $C[\cdot]$; I use Ψ to avoid a clash with consumption.) Then $Q = 1 + \Psi'$ (2.11), and the Euler equation is (2.12) with the CES marginal product. The cost vanishes at $I/K = g + d$, so on any balanced growth path $Q = 1$ and $F_K = r + d \equiv R$. Writing $F_K = (\alpha Y/K)^{1/\varepsilon}$:

$$\frac{K}{Y} = \alpha R^{-\varepsilon}, \quad \text{capital share} = \alpha R^{1-\varepsilon}, \quad \frac{W}{Z} = \left[\frac{1-\alpha R^{1-\varepsilon}}{1-\alpha}\right]^{\frac{1}{1-\varepsilon}}, \quad \frac{V}{Y} = (1+g)\,\alpha R^{-\varepsilon} \approx \frac{\alpha}{(r+d)^{\varepsilon}}. \qquad (1)$$

For $\varepsilon = 1$ this is (3.2), $V/Y = a/(r+d)$. The elasticity of value/output to the required return is $-\varepsilon$. The paper's argument is that production mutes the valuation effect of a fall in r, relative to the Gordon model, because dividends fall as investment rises. That muting is strongest when capital is a poor substitute for labor. Table 1 repeats the paper's Section 3 experiment (r from 7.5 to 4 percent, $g = 0$), normalizing α so the capital share is 0.282 initially. Pairing (i) gives closed forms on the balanced growth path for any ε. Away from it, CES makes capital's output elasticity vary with K/ZN, and the dynamics need numerical methods.

Table 1: Single capital stock, pairing (i): long-run effect of a fall in r from 7.5 to 4 percent ($g = 0$, $d = 0.066$).

	% change in V/Y	capital share, $r = 7.5\%$	capital share, $r = 4\%$
CES, $\varepsilon = 0.5$	15.3	0.282	0.245
CES, $\varepsilon = 1$	33.0	0.282	0.282
CES, $\varepsilon = 1.5$	53.4	0.282	0.325
Gordon model (dividends fixed)	87.5		

1.2 Pairing (ii): Cobb-Douglas production and Cobb-Douglas accumulation

Now let production be Cobb-Douglas without separate adjustment costs, and let capital accumulate through a Cobb-Douglas technology (Lucas and Prescott, 1971; Hercowitz and Sampson, 1991):

$$Y = K^{\alpha}(ZN)^{1-\alpha}, \qquad K(t+1) = \bar{A}\,K(t)^{1-\delta} I(t)^{\delta}, \qquad C + I = Y. \qquad (2)$$

Inverting the accumulation function shows that it *is* an adjustment-cost technology. The investment needed to reach K' is $I = K\,(K'/\bar{A}K)^{1/\delta}$, which is convex in the growth of capital. The price of installed capital is

$$Q = \frac{\partial I}{\partial K'} = \frac{I}{\delta K'}, \qquad \frac{\partial \ln(I/K)}{\partial \ln Q} = \frac{1}{1-\delta}.$$

To first order around a path with $I/K = g + d$, (2) is the usual $K' = (1-d)K + I$ when $\delta = (g+d)/(1+g)$ and $\bar{A} = (1+g)^{1-\delta}\delta^{-\delta}$, and then $Q = 1$ on that path. The quadratic cost gives an elasticity of I/K to Q of $1/(cx)$. Matching the two gives $c = (1-\delta)/x$. With the paper's $g = 0$ and $d = 0.066$, the CD technology implies $c \approx 14$, essentially the paper's "moderate" $c = 15$.

Proposition 1 (Exact solution, one capital stock)**.** *With $s = 1$ (so $U = b\ln C + (1-b)\ln L$) and any path for Z, anticipated or not, the equilibrium of (2) is*

$$\frac{I}{Y} = s_I \equiv \frac{B\delta\alpha}{1 - B(1-\delta)}, \qquad N = \frac{b(1-\alpha)}{b(1-\alpha) + (1-b)(1-s_I)}, \qquad \frac{V}{Y} = \frac{QK'}{Y} = \frac{s_I}{\delta} = \frac{\alpha}{\rho+\delta},$$

$$1 + r(t+1) = (1+\rho)\frac{Y(t+1)}{Y(t)}, \qquad W = \frac{(1-\alpha)Y}{N},$$

$$\ln K(t+1) = [1 - \delta(1-\alpha)]\ln K(t) + \delta(1-\alpha)\ln Z(t) + \textit{const.}$$

Proof sketch. Guess $V(K) = \kappa \ln K +$ terms in Z. The first-order condition $1/C = B\kappa\delta/I$ and the envelope condition $\kappa = \alpha Y/C + B\kappa(1-\delta)$ give constant I/Y and C/Y. The labor condition $(1-b)C/(bL) = (1-\alpha)Y/N$ then fixes N. Since C/Y is constant, $U_C(t)/U_C(t+1) = C(t+1)/C(t) = Y(t+1)/Y(t)$, which gives r. Finally, $V = QK' = I/\delta$.

Pairing (ii) makes the whole transition closed form, not just the balanced growth path. It also changes one of the paper's conclusions. Value/output is $\alpha/(\rho+\delta)$, the paper's (3.2) with the time-preference rate ρ in place of r. It is *independent of growth*: faster growth lowers K/Y but raises Q above one by exactly the offsetting amount. Under pairing (i), faster growth lowers V/Y. With $g = 0$, both pairings agree that a fall in the required return (a rise in B) raises V/Y by the same proportion as $\rho + \delta$ falls.

1.3 Pairing (ii), general case: CES production and CES accumulation

The Cobb-Douglas pair generalizes. Let $\rho \equiv (\varepsilon - 1)/\varepsilon$ and

$$Y^\rho = a_K K^\rho + a_L (ZN)^\rho, \qquad K(t+1)^\rho = \bar{A}^\rho\big[(1-\delta)K(t)^\rho + \delta I(t)^\rho\big], \qquad U = \sum_t B^t \frac{C_t^\rho}{\rho}, \tag{3}$$

with hours fixed. In the paper's preferences, fixed hours make utility a power of C with exponent $b(1-s)$, so the last condition is $b(1-s) = \rho$. Equivalently, the intertemporal elasticity of substitution $1/\theta$ equals the production elasticity ε. The accumulation technology is CES in old capital and new investment, with the same elasticity as production. As in the Cobb-Douglas case, inverting it gives a convex adjustment-cost function, $I = \{[(K'/\bar{A})^\rho - (1-\delta)K^\rho]/\delta\}^{1/\rho}$, and $Q = \partial I/\partial K' = (I/K')^{1/\varepsilon}/(\bar{A}^\rho\delta)$.

Proposition 2 (Exact solution, CES pair)**.** *For any path of Z, anticipated or not, the value*

function is $V_t(K) = \kappa K^\rho/\rho + h_t$, *and*

$$\frac{I}{Y} = s = \Lambda^\varepsilon, \qquad \Lambda \equiv \frac{B\bar{A}^\rho \delta a_K}{1 - B\bar{A}^\rho(1-\delta)}, \qquad \kappa = \frac{a_K(1-s)^{-1/\varepsilon}}{1 - B\bar{A}^\rho(1-\delta)},$$
$$K(t+1)^\rho = \bar{A}^\rho\big[(1-\delta+\delta a_K s^\rho)K(t)^\rho + \delta a_L s^\rho (Z(t)N)^\rho\big],$$
$$1 + r(t+1) = \frac{1}{B}\Big(\frac{Y(t+1)}{Y(t)}\Big)^{1/\varepsilon}, \qquad W = a_L Z\Big(\frac{Y}{ZN}\Big)^{1/\varepsilon},$$
$$\frac{WN}{Y} = a_L\Big(\frac{ZN}{Y}\Big)^\rho, \qquad \frac{V}{Y} = \frac{s^{1/\varepsilon}}{\bar{A}^\rho \delta}\Big(\frac{K'}{Y}\Big)^\rho.$$

As $\varepsilon \to 1$, $s \to B\delta\alpha/(1 - B(1-\delta))$ *and the law of motion becomes log-linear: Proposition 1.*

Proof sketch. With the guess for V_{t+1}, the Bellman objective is $(Y-I)^\rho/\rho + B\kappa\bar{A}^\rho[(1-\delta)K^\rho + \delta I^\rho]/\rho$. The first-order condition $C^{\rho-1} = B\kappa\bar{A}^\rho\delta I^{\rho-1}$ makes I/C, and therefore I/Y, a constant ϕ^ε with $\phi = B\kappa\bar{A}^\rho\delta$. Substituting back, the maximized objective is $(1+\phi^\varepsilon)^{1/\varepsilon}Y^\rho/\rho + B\kappa\bar{A}^\rho(1-\delta)K^\rho/\rho$. This is linear in K^ρ and $(ZN)^\rho$, so the guess reproduces itself with $\kappa = (1+\phi^\varepsilon)^{1/\varepsilon}a_K + B\bar{A}^\rho(1-\delta)\kappa$. Solving the two equations gives $s = \phi^\varepsilon/(1+\phi^\varepsilon) = \Lambda^\varepsilon$. The interest rate follows from $C \propto Y$ and the household Euler equation.

Three features matter for what follows. First, ρ appears in all three places (production, accumulation and utility). This is what makes the problem linear in K^ρ, and it is why CES production with Cobb-Douglas accumulation, or with log utility, is not exact. Second, the saving rate is Λ^ε, so $d\ln s = \varepsilon\, d\ln\Lambda$. Its response to the return on saving (through B, δ or a_K) is ε times the Cobb-Douglas response. Third, the normalization $\delta = x^{1/\varepsilon}/(x^{1/\varepsilon}+1-d)$ and $\bar{A}^\rho = (1+g)^{-1/\varepsilon}(x^{1/\varepsilon}+1-d)$, with $x = g + d$, makes the CES accumulation function agree to first order with $K' = (1-d)K + I$ on the balanced growth path. Then $Q = 1$ there, and the path coincides with pairing (i)'s. This generalizes the Cobb-Douglas normalization.

2 The model with AI capital

This section sets out the three-input model with pairing (i) technologies. Sections 4 and 5 replace the accumulation technology with pairing (ii): first Cobb-Douglas, then the general CES pair.

Technology. The traditional bundle and gross output are

$$X = K^a(ZN)^{1-a}, \qquad F = \Big[\omega^{1/\varepsilon}X^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)^{1/\varepsilon}M^{\frac{\varepsilon-1}{\varepsilon}}\Big]^{\frac{\varepsilon}{\varepsilon-1}}, \qquad M = \Gamma A. \tag{4}$$

Here A is AI capital measured in efficiency units and M is effective AI. With $\varepsilon < 1$, AI complements the capital-labor bundle; with $\varepsilon > 1$, it substitutes for it. The distribution weights carry the exponent $1/\varepsilon$, so $1-\omega$ is the AI income share when the effective cost of AI is one, and the $\varepsilon \to 1$ limit is $F = X^\omega M^{1-\omega}/[\omega^\omega(1-\omega)^{1-\omega}]$.

Two kinds of AI progress. *Disembodied*: Γ grows at rate γ_Γ and raises the productivity of the entire installed stock. *Embodied*: new AI efficiency units cost P_A in output, and P_A falls at rate γ_q, with $P_A(t)/P_A(t+1) = 1 + \gamma_q$. This is the investment-specific technical change of Greenwood, Hercowitz and Krusell (1997): a unit of output buys $1/P_A$ efficiency units, but units already installed do not improve.

Accumulation and adjustment costs. Following (2.7) for both stocks,

$$K(t+1) = I_K(t) + (1-d_K)K(t), \qquad A(t+1) = I_A(t) + (1-d_A)A(t), \tag{5}$$

$$Y = F - 0.5c_K(I_K/K - x_K)^2 K - P_A\, 0.5c_A(I_A/A - x_A)^2 A, \tag{6}$$

and $C + I_K + P_A I_A = Y$. AI adjustment costs are scaled by P_A, so they stay in proportion to AI spending as P_A falls, which is the appendix specification with its own investment price. The parameters x_K, x_A equal steady-state investment rates.

Optimality. Labor demand is $W = F_X(1-a)X/N$. Investment gives $Q_K = 1 + c_K(I_K/K - x_K)$ and $Q_A = P_A[1 + c_A(I_A/A - x_A)]$. Write $m_{t+1} = B\,U_C(t+1)/U_C(t) = 1/(1+r_{t+1})$. The Euler equations are

$$Q_K(t) = m_{t+1}\big\{F_K(t+1) - \Psi_K(t+1) + \Psi'_K(t+1)\, i_K(t+1) + (1-d_K)Q_K(t+1)\big\}, \tag{7}$$

$$Q_A(t) = m_{t+1}\big\{\Gamma(t+1)F_M(t+1) + P_A(t+1)[\Psi'_A\, i_A - \Psi_A](t+1) + (1-d_A)Q_A(t+1)\big\}, \tag{8}$$

where $i = I/K$ or I/A and $\Psi_j(i) = 0.5c_j(i - x_j)^2$. The first is (2.12). The second is its AI analogue. The household conditions are $(1-b)C/(bL) = W$ and $U_C(t) = B\,E_t[U_C(t+1)(1+r(t+1))]$. Firm value is $V(t) = Q_K(t)K(t+1) + Q_A(t)A(t+1)$; by Hayashi (1982) this equals the present value of dividends $D = Y - WN - I_K - P_A I_A$.

3 Pairing (i) with AI: closed forms on the balanced growth path

Define the *effective cost of AI*, the rental price of one unit of M,

$$\lambda \equiv \frac{P_A\, u_A}{\Gamma}, \qquad u_A = (1+r)(1+\gamma_q) - (1-d_A) \;\approx\; r + d_A + \gamma_q. \tag{9}$$

The user cost u_A contains an obsolescence term γ_q: a unit bought today loses value as newer, cheaper units arrive. The user cost of traditional capital is $R_K = r + d_K$. Let $\theta \equiv 1 - b(1-s)$ be the inverse intertemporal elasticity (footnote 11 of the paper).

Proposition 3 (Balanced growth)**.** *Suppose either (i) Γ/P_A is constant, for any ε, or (ii) $\varepsilon = 1$ with Γ growing at γ_Γ and P_A falling at γ_q. On the balanced growth path with growth g, $Q_K = 1$,*

$Q_A = P_A$, *hours are constant, and*

$$1 + r = (1+g)^{\theta}/B, \tag{10}$$

$$s_M = (1-\omega)\,\lambda^{1-\varepsilon} \quad \textit{(AI share of income)}, \tag{11}$$

$$\textit{labor share } \tfrac{WN}{Y} = (1-a)(1-s_M), \tag{12}$$

$$P_X = \big[(1-s_M)/\omega\big]^{\frac{1}{1-\varepsilon}} \ \ (\varepsilon \neq 1), \qquad P_X = \lambda^{-\frac{1-\omega}{\omega}} \ \ (\varepsilon = 1), \tag{13}$$

$$\frac{W}{Z} = (1-a)\Big[P_X\Big(\frac{a}{r+d_K}\Big)^{a}\Big]^{\frac{1}{1-a}}, \tag{14}$$

$$\frac{K}{Y} = \frac{a(1-s_M)}{r+d_K}, \qquad \frac{P_A A}{Y} = \frac{s_M}{u_A}, \tag{15}$$

$$\frac{V}{Y} = (1+g)\Big[\frac{K}{Y} + (1+\gamma_q)\frac{P_A A}{Y}\Big] \approx \frac{a(1-s_M)}{r+d_K} + \frac{s_M}{r+d_A+\gamma_q}, \tag{16}$$

$$N = \frac{b(1-a)(1-s_M)}{b(1-a)(1-s_M) + (1-b)\,C/Y}, \tag{17}$$

$$\frac{C}{Y} = 1 - (g+d_K)\frac{K}{Y} - \big[(1+g)(1+\gamma_q) - 1 + d_A\big]\frac{P_A A}{Y}. \tag{18}$$

Derivation. With Q's constant and adjustment costs zero, the Euler equations give $F_K = r + d_K$ and $\Gamma F_M = P_A u_A$, the latter because $Q_A = P_A$ falls at rate γ_q (a capital loss). Competitive pricing makes unit cost equal one, $1 = [\omega P_X^{1-\varepsilon} + (1-\omega)\lambda^{1-\varepsilon}]^{1/(1-\varepsilon)}$, where P_X is the shadow price of the bundle. This gives (11) and (13). The Cobb-Douglas bundle's cost function $P_X = (R_K/a)^a (W/[Z(1-a)])^{1-a}$, inverted, gives (14). The interest rate (10) is the household Euler equation with constant hours. Market value follows from $Q = 1$ and the timing of V.

Equation (16) is the AI generalization of (3.2): each capital stock is valued at its income share over its user cost. A compact way to read the wage formula is

$$\ln\frac{W}{Z} = \text{const} + \frac{\ln(1-s_M)}{(1-\varepsilon)(1-a)} - \frac{a}{1-a}\ln(r+d_K). \tag{19}$$

AI moves the real wage through two channels only: the share of income it absorbs (scaled by $1/(1-\varepsilon)$) and the interest rate. When $\varepsilon > 1$, a rising AI share raises the wage level even as it lowers the labor share. When $\varepsilon < 1$, cheaper AI shrinks its share and the wage gain is bounded.

3.1 Cobb-Douglas: an exact balanced growth path with permanent AI progress

Proposition 4 ($\varepsilon = 1$)**.** *With Γ growing at γ_Γ and P_A falling at γ_q,*

$$\ln(1+g) = \ln(1+g_Z) + \frac{1-\omega}{(1-a)\,\omega}\big[\ln(1+\gamma_\Gamma) + \ln(1+\gamma_q)\big], \tag{20}$$

AI capital in efficiency units grows at $(1+g)(1+\gamma_q)$, $s_M = 1-\omega$, the labor share is $(1-a)\omega$, and r follows from (10). The wage grows at g, and its level relative to Z is $W/Z = (1-a)\big[\lambda^{-\frac{1-\omega}{\omega}}(a/R_K)^a\big]^{\frac{1}{1-a}}$, with $\lambda = P_A u_A/\Gamma$ falling at $(1+\gamma_\Gamma)(1+\gamma_q)$.

Disembodied and embodied progress at the same rate deliver the same growth rate and the same real interest rate. They differ in levels, because embodied progress raises the AI user cost by the obsolescence term. Relative to disembodied progress at the same rate, the wage level is

Table 2: Cobb-Douglas balanced growth paths ($\varepsilon = 1$). Rates in percent. W/Z is relative to the no-progress path, with all three paths starting from the same Γ/P_A at $t = 0$.

	g	r	u_A	W/Z	WN/Y	K/Y	P_AA/Y	V/Y
No AI progress	1.50	7.50	27.5	1.000	0.696	1.94	0.109	2.08
Disembodied, $\gamma_\Gamma = 0.10$	1.92	7.94	27.9	0.987	0.696	1.88	0.107	2.03
Embodied, $\gamma_q = 0.10$	1.92	7.94	38.7	0.973	0.696	1.88	0.077	2.00

lower by $\frac{1-\omega}{(1-a)\omega}\ln[u_A^{\text{emb}}/u_A^{\text{dis}}]$ log points, P_AA/Y is lower by the factor $u_A^{\text{dis}}/u_A^{\text{emb}}$, and so is the AI contribution to V/Y. Table 2 shows this.

As in Kiley (2000), faster growth raises r one-for-one with $\theta = 1$ and *lowers* K/Y and V/Y. AI-driven growth does not mechanically raise valuation ratios. Embodied progress lowers V/Y further because AI assets depreciate economically as their price falls.

3.2 Substitutes ($\varepsilon > 1$): the AK regime

Proposition 5 ($\varepsilon > 1$)**.** *Let $\bar{A} \equiv (1-\omega)^{1/(\varepsilon-1)}$ and $r_Z = (1+g_Z)^\theta/B - 1$. With Γ/P_A constant, an interior balanced growth path exists if and only if $\lambda(r_Z) = P_A(r_Z + d_A)/\Gamma > \bar{A}$, i.e. $s_M < 1$. Otherwise the economy converges to an AK path in which $F \to \bar{A}\,\Gamma A$ and*

$$r = \bar{A}\,\Gamma/P_A - d_A, \qquad 1 + g = [B(1+r)]^{1/\theta} > 1 + g_Z. \tag{21}$$

On this path the bundle's shadow price grows at $\ln G_P = \frac{(1-a)(\ln(1+g)-\ln(1+g_Z))}{\varepsilon(1-a)+a}$, so

$$\Delta \ln W = \ln(1+g_Z) + \frac{\ln(1+g) - \ln(1+g_Z)}{\varepsilon(1-a)+a}, \qquad \Delta \ln \textit{(labor share)} = -(\varepsilon - 1)\ln G_P. \tag{22}$$

In the AK regime, the real wage grows faster than Z but more slowly than output, so the labor share declines toward zero while workers become richer. The interest rate is pinned by the AI rental rate, not by g_Z. If Γ/P_A keeps rising, so does the AK productivity $\bar{A}\Gamma/P_A$, and r and g rise without bound. This is the explosive-growth knife edge. At the baseline calibration with $\varepsilon = 1.5$, the threshold is reached only after Γ/P_A rises about 1,100-fold from its initial level, so the 20-fold shock studied below stays in the interior regime.

3.3 Complements ($\varepsilon < 1$): the Baumol limit

Proposition 6 ($\varepsilon < 1$)**.** *If $\Gamma/P_A \to \infty$, then $\lambda \to 0$, $s_M = (1-\omega)\lambda^{1-\varepsilon} \to 0$, $g \to g_Z$, $r \to r_Z$, the labor share $\to 1 - a$, $P_X \to \omega^{-1/(1-\varepsilon)}$, and*

$$\frac{W_\infty}{W_0} = (1 - s_{M,0})^{-\frac{1}{(1-\varepsilon)(1-a)}}. \tag{23}$$

AI becomes so cheap that it stops being a bottleneck. Output is limited by the traditional bundle, and the total wage gain from AI is capped by its initial income share. With $s_{M,0} = 0.03$, $\varepsilon = 0.5$ and $a = 0.282$, the cap is 8.9 percent.

4 Pairing (ii) with AI: exact dynamics

Set $\varepsilon = 1$, so that $F = K^{\alpha_K}(ZN)^{\alpha_N}(\Gamma A)^{\alpha_A}$ (up to a constant) with $\alpha_K = a\omega$, $\alpha_N = (1-a)\omega$, $\alpha_A = 1-\omega$. Replace both linear accumulation equations and both adjustment costs with Cobb-Douglas accumulation:

$$K(t+1) = \bar{A}_K K(t)^{1-\delta_K} I_K(t)^{\delta_K}, \qquad A(t+1) = \bar{A}_A A(t)^{1-\delta_A}\left[I_A(t)/P_A(t)\right]^{\delta_A}, \qquad C + I_K + I_A = Y. \tag{24}$$

Here I_A is AI spending in output units, so $1/P_A$ is the number of efficiency units one unit of output buys. Embodied progress (falling P_A) raises only the productivity of new AI investment. Disembodied progress (rising Γ) raises the productivity of the whole stock.

Proposition 7 (Exact solution with AI)**.** *With $s = 1$ and any paths of Z, Γ and P_A:*

1. *Saving rates are constant and separate by asset: $\dfrac{I_j}{Y} = s_j = \dfrac{B\delta_j\alpha_j}{1 - B(1-\delta_j)}$ for $j = K, A$. Hours are $N = b\alpha_N/[b\alpha_N + (1-b)(1-s_K-s_A)]$.*
2. *Market value is $V = Q_K K' + Q_A A' = I_K/\delta_K + I_A/\delta_A$, so $\dfrac{V}{Y} = \dfrac{\alpha_K}{\rho+\delta_K} + \dfrac{\alpha_A}{\rho+\delta_A}$, constant along every path.*
3. *$\ln(1+r_{t+1}) = \ln(1+\rho) + \Delta \ln Y_{t+1}$. The real wage is $W = \alpha_N Y/N$, so $\ln W$ moves one-for-one with $\ln Y$, and the labor share is constant at α_N.*
4. *With $x_t = (\ln K_t, \ln A_t)'$, $\alpha = (\alpha_K, \alpha_A)'$ and lowercase letters for logs,*

$$y_t = \alpha' x_t + \alpha_A \gamma_t + \alpha_N(z_t + \ln N),$$
$$x_{t+1} = M x_t + \begin{pmatrix}\delta_K \\ \delta_A\end{pmatrix}\left(\alpha_A\gamma_t + \alpha_N z_t\right) - \begin{pmatrix}0 \\ \delta_A\end{pmatrix} p_{A,t} + \text{const},$$
$$M = \begin{pmatrix}1-\delta_K(1-\alpha_K) & \delta_K\alpha_A \\ \delta_A\alpha_K & 1-\delta_A(1-\alpha_A)\end{pmatrix}.$$

   The eigenvalues of M are closed form ($\mu_{1,2} = \frac{1}{2}[\mathrm{tr}M \pm ((\mathrm{tr}\,M)^2 - 4\det M)^{1/2}]$). With $\delta_K = \delta_A = \delta$ they are $1-\delta$ and $1-\delta\alpha_N$.

The real interest rate and the real wage are therefore linear functionals of the AI technology paths:

$$\ln\frac{1+r_{t+1}}{1+r^*} = \alpha_A\Delta\gamma_{t+1} + \alpha'(x_{t+1} - x_t) \quad \text{(deviations from the no-shock path).} \tag{25}$$

Proposition 8 (Same area, different timing)**.** *Let $\ln\Gamma$ rise by Δ_Γ and $\ln(1/P_A)$ by Δ_q, along any paths. Then output rises in the long run by $\ln Y_\infty - \ln Y_\infty^* = \frac{\alpha_A}{\alpha_N}(\Delta_\Gamma + \Delta_q)$. The cumulative interest-rate deviation $\sum_{t\geq 0}\left[\ln(1+r_{t+1}) - \ln(1+r^*)\right]$ equals the same amount, less any jump in $\ln Y_0$ (a disembodied shock that arrives at $t = 0$ raises output on impact, and that windfall is not part of any ex-ante rate). The wage path is the output path. Embodied and disembodied progress of equal size move r by the same total and differ only in timing.*

For a surprise permanent step of size Δ at $t = 0$, (25) gives $\ln \frac{1+r_{t+1}}{1+r^*} = \alpha' M^t f$ with

$$f_{\text{dis}} = \alpha_A \Delta \begin{pmatrix} \delta_K \\ \delta_A \end{pmatrix}, \qquad f_{\text{emb}} = \Delta \begin{pmatrix} 0 \\ \delta_A \end{pmatrix}, \qquad \frac{\alpha' f_{\text{emb}}}{\alpha' f_{\text{dis}}} = \frac{\delta_A}{\alpha_K \delta_K + \alpha_A \delta_A}.$$

A disembodied step raises output immediately by $\alpha_A \Delta$. That gain is a *windfall* (an unexpected realized return), so it never shows up in the ex-ante real rate, and only the accumulation that follows raises r. An embodied step raises nothing on impact. All of its effect comes through investment, so the ex-ante real rate jumps by more: 9.6 versus 7.8 percent in the calibration below. When the same total improvement arrives gradually and is anticipated (the S-curve), the direct term $\alpha_A \Delta \gamma_{t+1}$ in (25) raises expected growth under disembodied progress. Then r rises sooner and more under disembodied progress, and the embodied response is a distributed lag of it with the same area.

Calibration and comparison with pairing (i). I set $\delta_j = (g_Z + d_j)/(1 + g_Z)$, which gives $\delta_K = 0.080$ and $\delta_A = 0.212$. The initial balanced growth path is then identical to the one in Section 6 (the saving rates equal $(g + d_j) \times$capital/output exactly). The eigenvalues of M are 0.943 and 0.794. Figure 1 and Table 3 compare the exact solution with the quadratic-cost model solved numerically, using the curvature that gives the same Q-elasticity ($c_K = 11.4$, $c_A = 3.7$). The two are almost indistinguishable. The exact model is therefore a reliable analytical guide to the quadratic-cost dynamics when $\varepsilon = 1$. Value/output is constant in the exact model and nearly constant in the quadratic one.

Table 3: 20-fold S-curve fall in the cost of AI, $\varepsilon = 1$: exact pairing (ii) versus pairing (i) with matched curvature.

	CD-CD (exact)		quadratic, matched c	
	disembodied	embodied	disembodied	embodied
peak r (pct.)	8.59	8.33	8.59	8.40
year of peak	10	13	10	13
wage gain at $t = 10$ (pct.)	4.8	1.8	4.8	1.8
wage gain at $t = 30$ (pct.)	12.4	11.7	12.4	11.8
V/Y range	2.080 (constant)		2.079–2.080	2.077–2.082

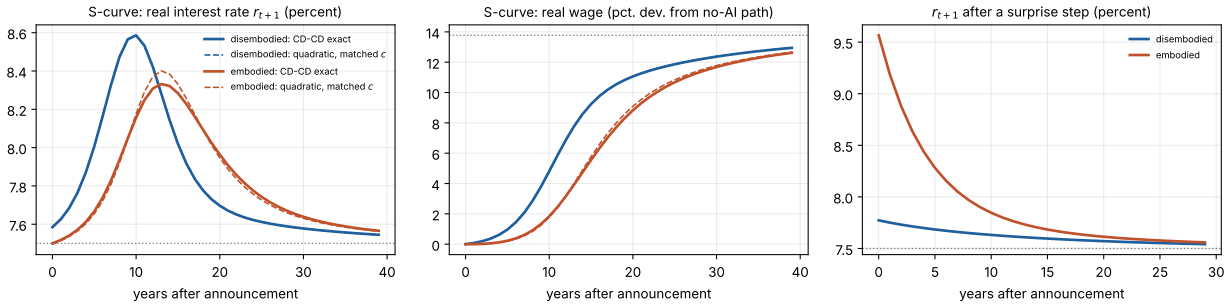


Figure 1: Pairing (ii), exact solution (solid) versus quadratic adjustment costs with matched curvature (dashed). Right panel: surprise permanent 20-fold step.

CES with Cobb-Douglas accumulation is not exact. The Cobb-Douglas pair depends on each input's output elasticity being constant. With $\varepsilon \neq 1$ production but Cobb-Douglas accumulation, AI's share $(1-\omega)\lambda^{1-\varepsilon}$ moves with the effective cost of AI, and so do the optimal saving rates. The script solves this mixed economy numerically (the disembodied S-curve). With $\varepsilon = 1.5$, I_A/Y rises from 0.030 to 0.105, and with $\varepsilon = 0.5$ it falls from 0.021 to 0.005. The exact solution for $\varepsilon \neq 1$ requires the CES accumulation function of the next section.

5 Pairing (ii) with AI, general case: CES production and CES accumulation

To keep the problem linear in X^ρ for every capital stock, all three inputs share the elasticity ε, and each capital stock accumulates through a CES function of the same elasticity:

$$\begin{gathered} Y^\rho = a_K K^\rho + a_L (ZN)^\rho + a_A (\Gamma A)^\rho, \\ K'^\rho = \bar{A}_K^\rho \left[(1-\delta_K)K^\rho + \delta_K I_K^\rho\right], \qquad A'^\rho = \bar{A}_A^\rho \left[(1-\delta_A)A^\rho + \delta_A (I_A/P_A)^\rho\right], \end{gathered} \tag{26}$$

with $U = \sum_t B^t C_t^\rho/\rho$ and hours fixed. AI is now equally substitutable for capital and labor. The nested form of Section 2, with a Cobb-Douglas K–L bundle inside the CES, is exact only when $\varepsilon = 1$, because $X^\rho = K^{a\rho}(ZN)^{(1-a)\rho}$ is not linear in K^ρ.

Proposition 9 (Exact solution with AI, CES pair)**.** *For any paths of Z, Γ and P_A, the value function is $V_t = (\kappa_{K,t} K^\rho + \kappa_{A,t} A^\rho)/\rho + h_t$. Define $\phi_{K,t} = B\kappa_{K,t+1}\bar{A}_K^\rho \delta_K$ and $\phi_{A,t} = B\kappa_{A,t+1}\bar{A}_A^\rho \delta_A P_{A,t}^{-\rho}$. Then*

$$\begin{gathered} \frac{I_{j,t}}{Y_t} = s_{j,t} = \frac{\phi_{j,t}^\varepsilon}{1 + \phi_{K,t}^\varepsilon + \phi_{A,t}^\varepsilon}, \qquad \Omega_t = \left(1 + \phi_{K,t}^\varepsilon + \phi_{A,t}^\varepsilon\right)^{1/\varepsilon}, \\ \kappa_{K,t} = \Omega_t a_K + B\bar{A}_K^\rho (1-\delta_K)\kappa_{K,t+1}, \qquad \kappa_{A,t} = \Omega_t a_A \Gamma_t^\rho + B\bar{A}_A^\rho (1-\delta_A)\kappa_{A,t+1}. \end{gathered}$$

Saving rates are functions of time alone. They are found by recursing backward from the final steady state and do not depend on the capital stocks. They do not depend on Z at all, and they are constant whenever Γ and P_A are. Prices are $1 + r_{t+1} = B^{-1}(C_{t+1}/C_t)^{1/\varepsilon}$ and $W = a_L Z (Y/ZN)^{1/\varepsilon}$. The labor share is $a_L(ZN/Y)^\rho$ and the AI share is $a_A(\Gamma A/Y)^\rho$.

Two features differ from the $\varepsilon = 1$ case. First, when $\varepsilon \neq 1$, news about AI changes saving rates on announcement, because $\kappa_{A,t}$ looks forward through Γ^ρ and $P_A^{-\rho}$. With $\varepsilon = 1$, saving rates ignore technology entirely. Second, the direction of that response depends on ε. With substitutes, cheaper future AI raises the value of AI capital, and I_A/Y rises. With complements, cheaper AI means less AI spending is needed, and I_A/Y falls.

Table 4: Exact solutions, CES pair with AI (IES = ε): 20-fold fall in the cost of AI. r_1 is the rate between years 0 and 1.

	I_A/Y		peak r (pct.), S-curve		r_1 (pct.), surprise step	
	initial	final	disemb.	emb.	disemb.	emb.
$\varepsilon = 0.5$	0.023	0.005	8.52	8.30	7.40	8.57
$\varepsilon = 1$	0.023	0.023	8.59	8.33	7.77	9.57
$\varepsilon = 1.5$	0.023	0.105	9.41	9.14	10.33	12.30

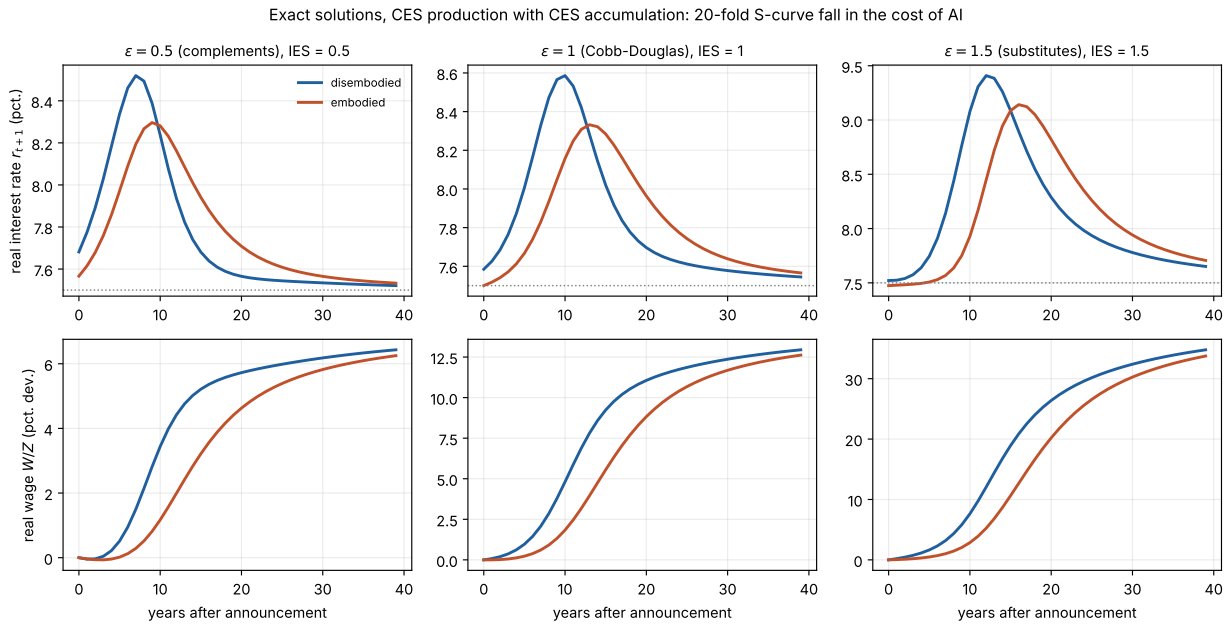


Figure 2: Exact solutions for the CES pair (Proposition 9; $\varepsilon = 1$ is Proposition 7). 20-fold S-curve fall in the cost of AI.

Results. The calibration has the same initial shares, interest rate and investment rates as before, with the IES set equal to ε (Table 4, Figure 2).

1. *S-curve.* For every ε, disembodied progress raises r sooner and higher than embodied progress (peaks of 8.5 vs 8.3 percent for $\varepsilon = 0.5$ and 9.4 vs 9.1 percent for $\varepsilon = 1.5$). Long-run wage gains are 6.8, 13.8 and 38 percent, close to the closed forms of pairing (i).

2. *Surprise step.* Embodied progress raises the ex-ante real rate more for every ε. With complements, a disembodied step *lowers* r, to 7.4 percent. The AI saving rate drops at once from 0.023 to 0.005, so consumption jumps up and then grows more slowly. With substitutes, the step raises r to 10.3 (disembodied) or 12.3 percent (embodied), because the AI investment boom pushes consumption growth up.

3. *Exactness depends on IES* $= \varepsilon$. With log utility and $\varepsilon = 0.5$, the generic solver shows that I_K/Y moves between 0.157 and 0.163 after a labor-technology shock, so it is no longer constant. With IES $= \varepsilon$, the same shock leaves both saving rates exactly unchanged.

6 Pairing (i) with AI: transitions for embodied versus disembodied shocks

The script `ai_model.py` solves the perfect-foresight path with a Newton method on the stacked equilibrium conditions, with all variables detrended by Z. The calibration is the paper's: $a = 0.282$, $b = 0.25$, $s = 1$, $d_K = 0.066$, $c_K = 15$, initial $r = 0.075$. Trend growth is $g_Z = 0.015$, where the paper used 0 for convenience. The AI block has $d_A = 0.20$, $c_A = 15$, an initial AI income share of 3 percent, and Γ_0 normalized so that $\lambda_0 = 1$, which makes the initial path identical for all ε. At $t = 0$

Table 5: 20-fold fall in the effective cost of AI: closed-form new BGP and transition statistics.

	s_M new BGP	WN/Y new BGP	W/Z new/old	V/Y new BGP	peak r (pct.) disemb.	emb.	W/Z at $t = 10$ disemb./emb.
$\varepsilon = 0.5$	0.007	0.713	1.068	2.04	8.18	7.93	1.039 / 1.014
$\varepsilon = 1$	0.030	0.696	1.138	2.08	8.59	8.22	1.048 / 1.015
$\varepsilon = 1.5$	0.134	0.622	1.372	2.25	9.66	8.93	1.061 / 1.015

agents learn that Γ/P_A will rise 20-fold along a logistic S-curve centered on year 10. The rise comes either through Γ (disembodied) or through P_A (embodied). Both shocks have the same initial and final balanced growth paths, which Proposition 3 gives in closed form, so all of the difference lies in the transition.

Results.

1. *The real interest rate rises temporarily in both cases, and more under disembodied progress.* Disembodied progress raises the productivity of AI already in place, so output and expected consumption growth rise as the S-curve unfolds, and r rises with them. The peak is 8.6 percent in the Cobb-Douglas case and 9.7 percent with $\varepsilon = 1.5$. Embodied progress pays off only through new vintages, and accumulation is slowed by adjustment costs. The rise in r is therefore smaller (8.2 and 8.9 percent at the peak), comes later and lasts longer. Under complements the rise is smallest, because AI's share collapses as it becomes cheap. In every case r returns to r_Z: a one-time level shift has no long-run effect on r. Only a permanent change in the growth rate does (Table 2).

2. *The real wage rises in all cases, by an amount set by ε.* The long-run gain is 6.8 percent with complements (close to the 8.9 percent Baumol cap), 13.8 percent with Cobb-Douglas, and 37 percent with substitutes, all from (19). Under embodied progress the wage lags by roughly five years: ten years in, the wage is up 1.5 percent under embodied progress versus 4–6 percent under disembodied progress.

3. *Labor share.* The gross labor share moves only through s_M. It rises with complements and falls with substitutes, from 69.7 to 62.2 percent at $\varepsilon = 1.5$. In the Cobb-Douglas case it is constant gross of adjustment costs. The temporary rise under embodied progress reflects the adjustment costs of the AI investment boom, which reduce net output.

4. *Valuation.* Embodied progress generates an AI investment boom with $Q_A/P_A > 1$ and a temporary rise in V/Y. Disembodied progress in the complements case *lowers* V/Y in the transition, as in the paper's growth experiment. In the long run, V/Y follows (16) and rises only when the AI share rises ($\varepsilon > 1$).

7 Verification

The script checks every closed form against the equilibrium conditions directly. It builds level paths, without detrending, from the closed forms and evaluates the production function, both Euler equations, the labor condition and the household Euler equation. It then compares V with the present value of dividends. It does this for the Cobb-Douglas path with permanent embodied

and/or disembodied progress and for CES steady states with $\varepsilon \in \{0.5, 1.5\}$. It also checks the Baumol limit, the consistency of the AK threshold, and that every transition converges to the closed-form new path. Table 6 reports the largest error for each check.

Table 6: Verification, pairing (i) (`ai_model.py`): maximum absolute (or relative) errors.

Check	max abs. error
CD BGP: F = Y (prod. fn. at closed form)	6.2e-15
CD BGP: labor FOC (1-b)C/(bL) = W	1.6e-15
CD BGP: Euler K (2.12)	9.8e-15
CD BGP: Euler A	1.0e-14
CD BGP: r = (1+g)$^t heta/B - 1$	2.2e-15
CD BGP: wage W/Z closed form	3.3e-15
CD BGP: labor share	2.2e-16
CD BGP: V/Y closed form	1.6e-14
CD BGP: V = PV(dividends) (Hayashi)	1.9e-14
CES steady state: F = Y (prod. fn. at closed form)	8.9e-16
CES steady state: labor FOC (1-b)C/(bL) = W	1.0e-15
CES steady state: Euler K (2.12)	1.2e-14
CES steady state: Euler A	1.0e-14
CES steady state: r = (1+g)$^t heta/B - 1$	2.2e-15
CES steady state: wage W/Z closed form	8.9e-16
CES steady state: labor share	4.4e-16
CES steady state: V/Y closed form	1.2e-14
CES steady state: V = PV(dividends) (Hayashi)	1.2e-14
Baumol limit W/Z (e=0.5)	8.4e-08
AK threshold consistent (e=1.5)	0.0e+00
AK threshold: g AK - g Z at threshold	9.7e-17
transition end W/Z vs closed form, e=0.5, disembodied	2.0e-05
transition end W/Z vs closed form, e=0.5, embodied	3.2e-05
transition end W/Z vs closed form, e=1.0, disembodied	5.3e-05
transition end W/Z vs closed form, e=1.0, embodied	8.8e-05
transition end W/Z vs closed form, e=1.5, disembodied	3.1e-04
transition end W/Z vs closed form, e=1.5, embodied	4.9e-04
transition Newton residual (max)	3.7e-12

For pairing (ii), `cdcd_model.py` solves the nonlinear perfect-foresight equilibrium with a Newton method that does not impose constant saving rates, starting from saving rates 10 percent away from the closed form. It confirms that the solution has constant saving rates and hours, the closed-form r path, constant V/Y, the analytic step responses $\alpha' M^t f$, Proposition 8, and the exact growth rate under permanent progress (Table 7).

For the CES pair, `ces_exact.py` solves the nonlinear equilibrium (two Euler equations with the CES accumulation technology) starting from saving rates 5 percent away from the closed form. It confirms the backward-recursion saving rates for both elasticities, both kinds of progress, and both S-curve and step shocks. It also checks $s = \Lambda^{\varepsilon}$, the match with the balanced growth path of pairing (i), and the invariance of saving rates to Z (Table 8).

8 Extensions

- Calibrating ε, d_A and γ_q to AI and software investment data, and comparing the implied paths for r with long-term real yields.
- Letting AI substitute for labor within the bundle (nested CES with task structure) to separate wage-level from labor-share effects.

Table 7: Verification, pairing (ii) (`cdcd_model.py`).

Check	max abs. error
CD-CD saving rates = quadratic-AC BGP	1.1e-16
CD-CD hours = quadratic-AC BGP	0.0e+00
V/Y: sum s/delta = sum alpha/(rho+delta)	1.8e-15
CD-CD disembodied: generic solve $s_K, s_A constant$	1.0e-14
CD-CD disembodied: generic solve hours	2.8e-15
CD-CD disembodied: r path = (1+rho)Y'/Y - 1	2.4e-15
CD-CD disembodied: V/Y constant	1.4e-13
CD-CD disembodied: ln Y gain at T=120 vs (aA/aN) Delta	6.3e-05
CD-CD disembodied: sum of ln(1+r) deviations = ln Y gain	1.4e-15
CD-CD embodied: generic solve $s_K, s_A constant$	1.0e-14
CD-CD embodied: generic solve hours	2.8e-15
CD-CD embodied: r path = (1+rho)Y'/Y - 1	2.7e-15
CD-CD embodied: V/Y constant	1.4e-13
CD-CD embodied: ln Y gain at T=120 vs (aA/aN) Delta	8.6e-05
CD-CD embodied: sum of ln(1+r) deviations = ln Y gain	5.6e-17
CD-CD permanent disembodied progress: Newton residual at closed form	4.4e-16
CD-CD permanent disembodied progress: growth at T=120 vs closed form	3.4e-07
CD-CD permanent embodied progress: Newton residual at closed form	6.7e-16
CD-CD permanent embodied progress: growth at T=120 vs closed form	4.7e-07
step disembodied: ln(1+r) dev = alpha' $M^t f$	4.6e-16
step embodied: ln(1+r) dev = alpha' $M^t f$	6.1e-16

- Adding an equity premium as in Section 3 of the paper, by moving B, to separate growth-driven from discount-rate-driven valuation.

References

Greenwood, J., Z. Hercowitz and P. Krusell (1997). "Long-Run Implications of Investment-Specific Technological Change." *American Economic Review* 87(3): 342–362.

Hercowitz, Z. and M. Sampson (1991). "Output Growth, the Real Wage, and Employment Fluctuations." *American Economic Review* 81(5): 1215–1237.

Hayashi, F. (1982). "Tobin's Marginal q and Average q: A Neoclassical Interpretation." *Econometrica* 50: 213–224.

Kiley, M. T. (2000). "Stock Prices and Fundamentals in a Production Economy." FEDS Working Paper 2000-05, Federal Reserve Board.

Kiley, M. T. (2004). "Stock Prices and Fundamentals: A Macroeconomic Perspective." *Journal of Business* 77(4): 909–936.

Lucas, R. E. and E. C. Prescott (1971). "Investment under Uncertainty." *Econometrica* 39(5): 659–681.

Table 8: Verification, CES pair (`ces_exact.py`).

Check	max abs. error
e=0.5: steady $s_K = (g + d_K)K/Y$	2.2e-16
e=0.5: steady $s_A = (g + d_A)A/Y$	1.0e-17
e=0.5: one-capital s = Lambda^e	1.4e-16
e=0.5 disembodied S-curve: generic solve = closed-form saving rates	1.1e-16
e=0.5 disembodied step: generic solve = closed-form saving rates	1.7e-15
e=0.5 embodied S-curve: generic solve = closed-form saving rates	8.6e-15
e=0.5 embodied step: generic solve = closed-form saving rates	1.7e-15
e=0.5: labor-technology shock leaves $s_K, s_A constant$	1.1e-16
e=0.5: and confirmed by generic solve	1.1e-15
e=1.5: steady $s_K = (g + d_K)K/Y$	6.7e-16
e=1.5: steady $s_A = (g + d_A)A/Y$	1.7e-17
e=1.5: one-capital s = Lambda^e	9.4e-16
e=1.5 disembodied S-curve: generic solve = closed-form saving rates	1.3e-13
e=1.5 disembodied step: generic solve = closed-form saving rates	1.1e-12
e=1.5 embodied S-curve: generic solve = closed-form saving rates	9.4e-14
e=1.5 embodied step: generic solve = closed-form saving rates	1.1e-12
e=1.5: labor-technology shock leaves $s_K, s_A constant$	5.7e-16
e=1.5: and confirmed by generic solve	9.8e-13

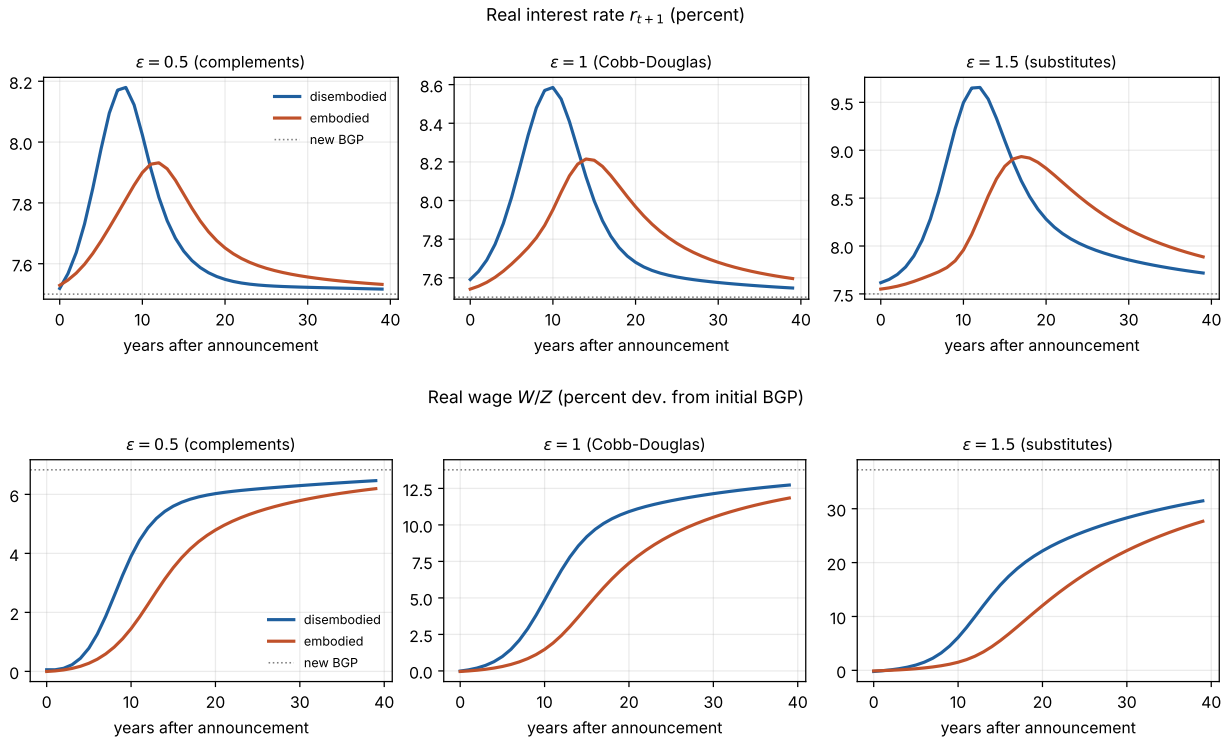


Figure 3: Real interest rate and real wage after an S-curve AI shock. Dotted lines: closed-form new BGP.

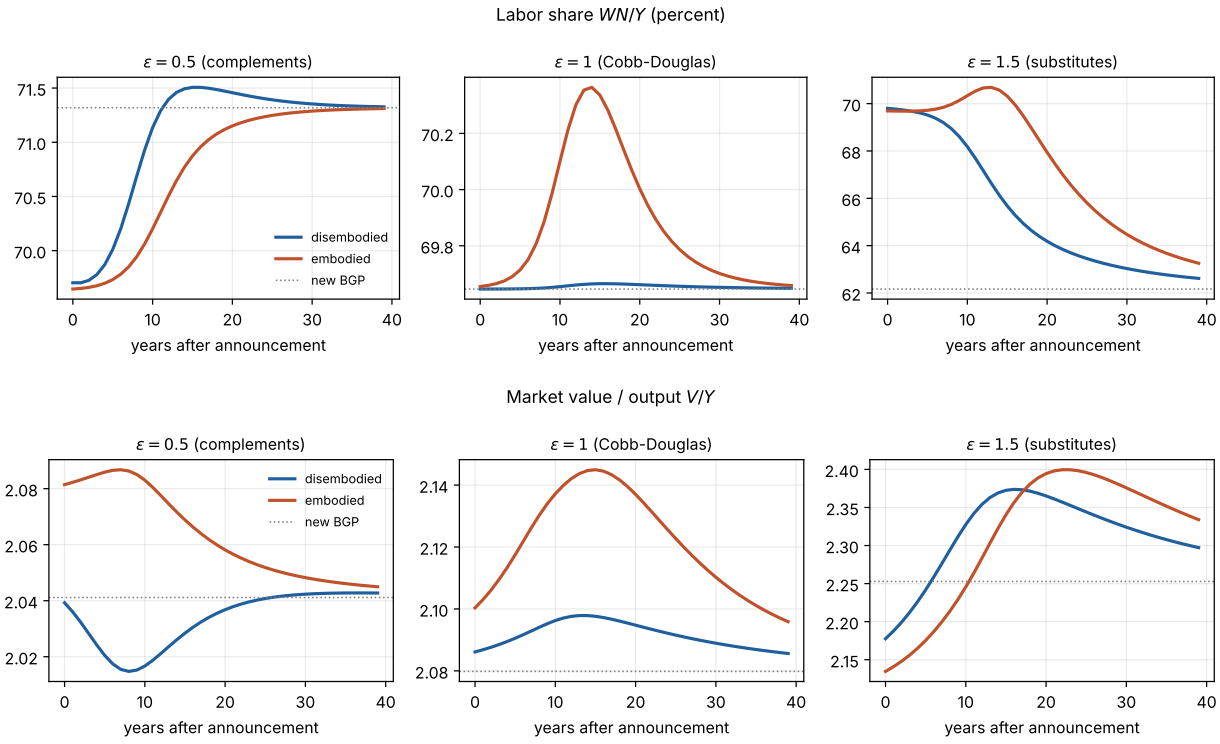


Figure 4: Labor share (relative to output net of adjustment costs) and market value relative to output.